SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	CSE-AI	**Date:**	29/07/26

Code of Conduct

*I **A. Mounish / 192472273** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ___________________________

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3

Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as "analyzing", "analysis", and "analytical". The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

2. An AI-based sentiment analysis platform receives product reviews containing words like "disagree", "agreement", and "agreeable". The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as "govern", "government", and "governance". The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

4. A document classification system processes input words such as "activate", "activation", and "reactivation" to improve semantic grouping and indexing.
Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.
Analyze how prefix addition and suffix transformation affect meaning and word class.
Apply the process to decompose each word and map them to a unified base representation. Derive the morphological structure and normalized root for each input word.

5. A search optimization module processes input words such as "create", "creates", and "creating" to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.
Identify suffix patterns and map all variations to a common base form without altering meaning.
Apply the process to extract root forms and classify each transformation. Determine the morphological breakdown and normalized output for each word.

Q1. Morphological Processing Approach

Scenario

A search engine must group analyzing, analysis, analytical into one common representation for better indexing.

Step 1: Morphological Decomposition

Word	Root	Affix
analyzing	analyze	-ing
analysis	analyze	-sis
analytical	analyze	-ical

Step 2: Identify Forms

Word	Type
analyzing	Inflectional
analysis	Derivational
analytical	Derivational

Step 3: Normalization

Common base form = analyze

Morphological Structure

- **analyzing = analyze + ing**
- **analysis = analyze + sis**
- **analytical = analyze + ical**

Final Normalized Output

Input Word	Normalized Form
analyzing	analyze
analysis	analyze
analytical	analyze

Q2. Morphological Parsing Strategy

Scenario

A sentiment analysis system processes disagree, agreement, agreeable.

Step 1: Morphological Parsing

Word	Prefix	Base Form	Suffix
disagree	dis-	agree	—
agreement	—	agree	-ment
agreeable	—	agree	-able

Step 2: Semantic Effect

- **dis- → Negation**
- **-ment → Forms noun**
- **-able → Forms adjective**

Step 3: Classification

Word	Transformation
disagree	Derivational
agreement	Derivational
agreeable	Derivational

Root Extraction

Common root = agree

Final Output

Input	Root
disagree	agree
agreement	agree
agreeable	agree

Q3. Morphology-Based Normalization

Scenario

Normalize govern, government, governance for topic modelling.

Step 1: Decomposition

Word	Root	Suffix
govern	govern	—
government	govern	-ment
governance	govern	-ance

Step 2: Derivational Layers

- **govern → Verb**
- **government → Noun**
- **governance → Abstract noun**

Step 3: Common Base Form

govern

Morphological Breakdown

- **govern = govern**
- **government = govern + ment**
- **governance = govern + ance**

Final Output

Input	Normalized Form
govern	govern
government	govern
governance	govern

Q4. Morphological Parsing Strategy

Scenario

Analyze activate, activation, reactivation.

Step 1: Decomposition

Word	Prefix	Root	Suffix
activate	—	activate	—
activation	—	activate	-ion
reactivation	re-	activate	-ion

Step 2: Derivational Layers

- **activate → Verb**
- **activation → Noun**
- **reactivation → Prefix + Noun**

Step 3: Semantic Changes

- **re- → Again**
- **-ion → Converts verb into noun**

Step 4: Unified Base Representation

Common root = activate

Morphological Structure

- **activate = activate**
- **activation = activate + ion**
- **reactivation = re + activate + ion**

Final Output

Input	Normalized Root
activate	activate
activation	activate
reactivation	activate

Q5. Morphology-Based Normalization

Scenario

Normalize create, creates, creating.

Step 1: Identify Suffix Pattern

Word	Root	Suffix
create	create	—
creates	create	-s
creating	create	-ing

Step 2: Classification

Word	Transformation
create	Base Form
creates	Inflectional (Present Tense)
creating	Inflectional (Present Participle)

Step 3: Common Base Form

create

Morphological Breakdown

- **create = create**
- **creates = create + s**
- **creating = create + ing**

Final Normalized Output

Input	Normalized Form
create	create
creates	create
creating	create

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario

Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	**Understanding of Scenario**	**Identification of Key Issues**	**Application of Concepts**	**Decision Making**	**Clarity of Response**	**Sub. Total**	**Total %**
1							
2							
3							
4							
5							

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________